

UTF-8 validation without processor lock-in

Two Parabix expressions · exact-
boundary correctness · performance
study

[Team member names] · August 5, 2026

UTF-8 is locally encoded—but globally stateful

ASCII

00–7F stands alone. It is the fast path for most English text.

Continuation

80–BF is legal only where a prior leader created scope.

Leaders

C2–DF, E0–EF, and F0–F4 require one, two, or three following bytes.

Narrow ranges

E0, ED, F0, and F4 constrain byte two to prevent overlong, surrogate, or out-of-range values.

Three implementations isolate the compiler trade-off



DIRECT IDISA
Packed byte lanes
comparisons · dslli · XOR
· popcount



PABLO / u8u16
Basis bit streams
character classes ·
Advance · AtEOF



KEISER-LEMIRE
Native SIMD vectors
lookup/shuffle-oriented
hand-written code

The same UTF-8 rule becomes two compiler-facing programs

Direct bytes

Keep the previous vector block. Align bytes at distances 1/2/3. XOR expected continuation lanes with actual 80–BF lanes. Add narrow range masks. Reduce error lanes.

Pablo scopes

Transpose bytes into eight basis streams. Compile byte classes. Advance leaders into scope22/32/33/42/43/44. XOR scope with suffixes. Mark outstanding scope at EOF.

Exact block boundaries exposed a real correctness gap

Original behavior

... 41 41 E2 | EOF

A partial block supplied zeros and often exposed missing suffixes. At an exact multiple, no following padded block was guaranteed. The unfinished leader could be accepted.

Completed behavior

Keep direct state across blocks.

Add a direct-byte EOF guard.

Count Add1 error streams correctly.

AtEOF(outstanding scope) makes termination part of validation.

Both compiler-generated validators agree with the oracle

124 generated cases per width · malformed ranges · crossing sequences · $B \pm 3$ · exact B/2B/3B truncation

Oracle / implementation	64-bit	128-bit	256-bit	512-bit	Overall
Python strict UTF-8	124/124	124/124	124/124	124/124	496/496
Direct IDISA + EOF guard	124/124	124/124	124/124	124/124	496/496
Pablo validation-only	124/124	124/124	124/124	124/124	496/496

496 / 496 cases passed

Direct · Pablo · independent strict decoder

Performance evidence is the remaining group checkpoint

Do not fill this slide with invented throughput

The groupmate must insert the verified single-thread chart and 1/2/4/8-thread scaling chart from raw CSV before submission.

10+

measured trials
after warmup

1 · 2 · 4 · 8

threads, capped
at physical cores

GB/s + IQR

median throughput
and variability

WHAT WE
LEARNED

EOF belongs in the algorithm

Portable SIMD works.
Boundaries are semantic.
Next: benchmark
codegen.